

COMMUTE SERVICE: PLATFORM BLUEPRINT & OPERATIONAL ROADMAP

Document Version: 1.0 (Pre-Build / Seed Stage)

Target Market: Gated Estates & High-Density Commuter Corridors (Abuja, Nigeria)

Proposed Seed Capital: ₦5,000,000 for 5% Equity (Valuation: ₦100,000,000)

1. Executive Summary & First Principles Foundations

Commute Service is a hyper-local, trust-driven micro-mobility and closed-loop rideshare framework. It solves the critical problem of skyrocketing transit costs and security vulnerabilities for daily commuters moving from residential districts (e.g., Lugbe) to urban employment hubs (e.g., CBD, Wuse, Garki).

By matching verified resident drivers who have empty car seats with verified neighbors heading along the exact same route, the platform creates an asset-light, highly defensible transport network. Instead of building high-friction native mobile applications, Commute Service utilizes an optimized mobile web frontend paired with automated WhatsApp notification engines, maximizing user adoption and minimizing engineering overhead.

2. Technical Architecture & Core Features

A. The Core Tech Stack

- **Database & Auth:** Supabase (PostgreSQL) handling real-time data sync, state machinery, and Row-Level Security (RLS).
- **Hosting & API Layer:** Vercel edge networks running Node.js/TypeScript serverless functions.
- **Communications Interface:** WhatsApp Business API webhooks (via Twilio or dedicated gateway) for real-time match pings.

B. Core Features

1. Multi-Tier Identity Layer:

- *Tier 1 (Gated Estate):* Cross-references users with pre-approved Homeowners Association (HOA) registries or verified estate access credentials.
- *Tier 2 (Corporate Anchor):* One-Time Password (OTP) verification via official corporate email domains (e.g., .gov.ng, corporate company extensions) to capture non-gated professional commuters.
- *Tier 3 (Social Voucher):* A decentralized peer-vouching network where Tier 1/2 users can issue single-use signup invites to trusted neighbors.
- *Tier 4 (Digital KYC):* Integration with automated verification APIs (e.g., Smile ID) for National Identification Number (NIN) and live biometric face-matching.

2. Predictive Route Matcher:

- Algorithmic grouping of drivers and riders based on static, recurring weekly schedules (e.g., Monday–Friday, departing at 7:15 AM).
- Real-time seat inventory counters tracking available space per vehicle.

3. The Micro-Transaction Escrow Ledger:

- Pre-funded internal user virtual wallets to eliminate erratic payment gateway failures during live transits.
- Atomic database functions that lock transaction fees (₦50 platform fee + driver's cost-split) in escrow at booking time and release them automatically upon safe ride completion.

3. End-to-End User Experience & Data Flow

[User Onboarding: QR Scan / Domain Link] → [Trust Engine Verification (Tier 1-4)]



[Automated Escrow Lock] ← [One-Click Booking] ← [WhatsApp Route Notification]



[Ride Completed] → [Split Released to Driver Wallet] + [₦50 Platform Fee Collected]

Step 1: Verification & Activation

- **Gated Estates:** The resident scans a location-specific QR code displayed at estate exit gates or recreation points. They input their profile, which is approved via an automated database match or an on-the-ground Community Ambassador's administrative console.
- **Non-Gated Areas:** The resident signs up via a corporate email verification link or inputs their NIN for instant biometric validation.

Step 2: Route Listing & Matching

- On Sunday, a driver lists their standard weekly commute timeline.
- The PostgreSQL backend automatically clusters matching rider profiles living along the same geographic path.
- The system pushes an automated WhatsApp message to the matching riders with a direct link to lock down a seat.

Step 3: Transit Execution & Settlement

- The rider accepts the seat via the web view interface. Funds move into an escrow state.
- The driver picks up passengers inside the secure perimeter or designated neighborhood hub.

- Upon arrival at the central city destination, the driver marks the trip completed. Escrow releases the fuel-split contribution to the driver's wallet and moves the ₦50 fee to the corporate account.

4. Hyper-Local Marketing, Ads, & Promotion Strategy

To keep customer acquisition costs near zero, marketing avoids broad, wasteful advertising and focuses heavily on high-density conversions.

A. Gated Estate Rollout (Organic Trust)

- **The Community Ambassador Program:** Appointing well-connected estate residents or estate executive members to act as local network drivers. They receive a performance-linked micro-fee (e.g., ₦10 per transaction) for keeping their specific estate active.
- **Physical Choke-Point Assets:** Placing high-visibility roll-up banners and gate flyers featuring clear QR codes directly at the estate's primary security check-points.

B. Non-Gated Market Scale (Strategic Node Capturing)

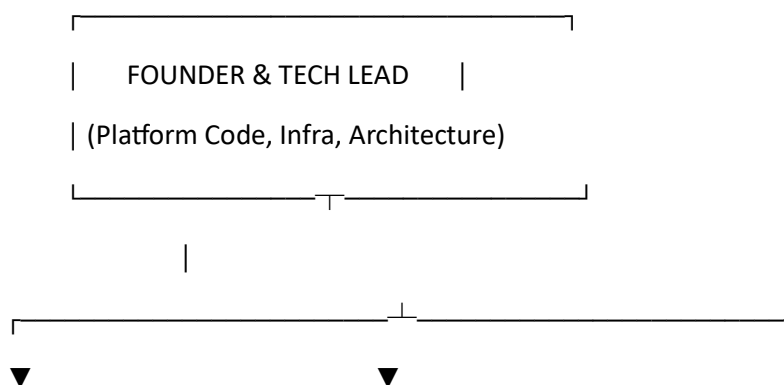
- **B2B Corporate Infiltration:** Pitching the platform directly to staff cooperatives, tech hubs, and government ministries in the CBD/Wuse as a structured staff transit benefit scheme.
- **Geofenced Digital Campaigns:** Running highly localized Meta (Facebook/Instagram) ads restricted to a tight 2-kilometer radius around high-traffic junctions (e.g., Lugbe Total station, Federal Housing) active exclusively between 6:30 AM and 8:30 AM.

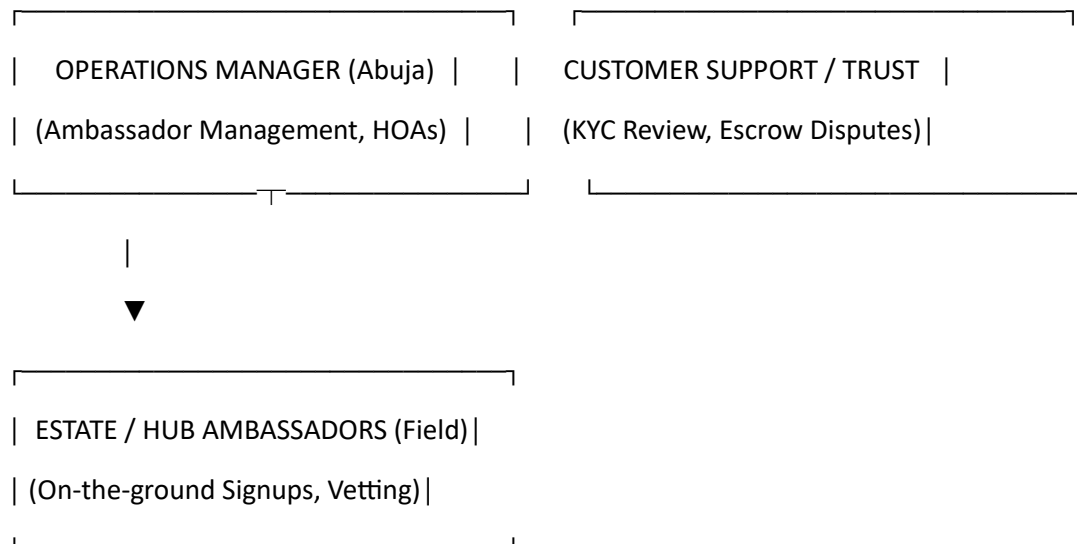
C. Promotional Launch Campaigns

- **The "First 3 Rides Free" Initiative:** Fully subsidizing the initial platform fees and cost-splits for new riders out of the operational runway to establish a recurring commute habit.
- **Driver Earnings Guarantee:** Providing a baseline daily fuel subsidy (e.g., ₦1,500/day) during launch weeks to early drivers who publish their routes, ensuring system availability even before passenger density peaks.

5. Organizational & Admin Structure

To remain lean, the team operates on a dual-track model separating core platform engineering from localized field operations.





1. Founder & Technical Lead (Your Role)

- **Responsibilities:** Managing the complete application build, infrastructure deployments (Supabase/Vercel), API routing, database maintenance, framework optimizations, and architectural security policies.

2. Operations Manager (Abuja-Based)

- **Responsibilities:** Handling physical corporate outreach to ministries and businesses, negotiating platform access with Estate Homeowners Associations (HOAs), managing the distribution of physical marketing assets, and leading the Community Ambassador network.

3. Estate / Community Hub Ambassadors (Field Team)

- **Responsibilities:** Managing localized signups, executing manual validation checks for estate whitelist overrides, distributing flyers at security gates, and serving as the first line of community trust.

4. Customer Support & Trust Officer (Remote)

- **Responsibilities:** Auditing Tier 4 KYC document exceptions, resolving wallet or escrow settlement disputes, and handling incoming driver/rider feedback flags.

6. Financial Budget Allocation (₦5,000,000 Seed Runway)

This allocation ensures that 100% of the cash injection is spent on operational scaling and market acquisition, leveraging the fact that the platform software development is executed entirely in-house without outside agency fees.

- **Product Growth & User Subsidies (50% / ₦2,500,000):** Allocating direct funds toward the "First 3 Rides Free" campaigns and driver baseline guarantees to force liquid transaction density in early hubs.
- **Operations & Ambassador Retainers (20% / ₦1,000,000):** Performance-based milestones and activation stipends for local estate and hub managers.

- **Geofenced Digital Ads & B2B Outreach (20% / ₦1,000,000):** Targeted social media ad spend around transit chokepoints and corporate marketing pipelines.
- **Physical Print & Infrastructure Assets (10% / ₦500,000):** Estate gate banners, QR-code posters, verification API integrations (Smile ID/Mono tokens), and cloud database runtime costs.